



Dyspnea

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Dyspnea refers to unpleasant or uncomfortable breathing. It is experienced and described differently by patients depending on the cause.

Etiology of Dyspnea

Dyspnea has many pulmonary, cardiac, and other causes (1), which vary by acuity of onset (see tables [Some Causes of Acute Dyspnea](#), [Some Causes of Subacute Dyspnea](#), and [Some Causes of Chronic Dyspnea](#)).

The **most common** causes include:

- [Asthma](#)
- [Chronic obstructive pulmonary disease](#) (COPD)
- [Heart failure](#)
- [Myocardial ischemia](#)
- Physical deconditioning
- [Pneumonia](#)

The most common cause of dyspnea in patients with chronic pulmonary or cardiac disorders is:

- Exacerbation of their disease

However, such patients may also acutely develop another condition (eg, a patient with long-standing asthma may have a myocardial infarction, a patient with chronic heart failure may develop pneumonia).

TABLE		
Some Causes of Acute* Dyspnea		
Cause	Suggestive Findings	Diagnostic Approach †
Pulmonary causes		
Asthma , bronchospasm, or reactive airway disease	Wheezing and poor air exchange that arise	History and physical examination

Cause	Suggestive Findings	Diagnostic Approach †
	spontaneously or after exposure to specific stimuli (eg, allergen, URI, cold, exercise) Possibly pulsus paradoxus Often a preexisting history of reactive airway disease	Sometimes pulmonary function testing or peak flow measurement
<u>Foreign body inhalation</u>	Sudden onset of cough or stridor in a patient (typically an infant or young child) without URI or constitutional symptoms	Inspiratory and expiratory chest radiographs Sometimes bronchoscopy
<u>Pneumothorax</u>	Abrupt onset of sharp chest pain, tachypnea, diminished breath sounds, and hyperresonance to percussion May follow injury or occur spontaneously (especially in tall, thin patients and in patients with COPD)	Chest radiograph
<u>Pulmonary embolism</u>	Abrupt onset of sharp chest pain, tachypnea, and tachycardia Often risk factors for pulmonary embolism (eg, cancer, immobilization, DVT, pregnancy, use of oral contraceptives or other estrogen-containing medications, recent surgery or hospitalization, family history)	CT angiography Less often, V/Q scanning and possibly pulmonary arteriography
<u>Toxin-induced airway damage</u> (eg, due to inhalation of chlorine or hydrogen sulfide)	Sudden onset after occupational exposure or inappropriate use of cleaning agents	Inhalation usually obvious by history Chest radiograph

Cause	Suggestive Findings	Diagnostic Approach †
		Sometimes ABG measurement and observation to determine severity
Cardiac causes		
<u>Acute myocardial ischemia or infarction</u>	Substernal chest pressure or pain that may or may not radiate to the arm or jaw, particularly in patients with risk factors for CAD	ECG Cardiac enzyme testing
Papillary muscle dysfunction or rupture	Sudden onset of chest pain, new or loud holosystolic murmur, and signs of heart failure, particularly in patients with recent MI	History and physical examination Echocardiography
Other causes		
Anxiety disorder causing hyperventilation	Situational dyspnea often accompanied by psychomotor agitation and paresthesias in the fingers or around the mouth Normal examination findings and pulse oximetry measurements	History and physical examination Diagnosis of exclusion
Acute diaphragmatic paralysis	Often sudden onset after trauma affecting the phrenic nerve‡ Frequent orthopnea	Chest radiograph Fluoroscopic sniff test

* Acute dyspnea occurs within minutes of triggering event.

† Most patients should have pulse oximetry and, unless symptoms are clearly a mild exacerbation of known chronic disease, chest radiograph.

‡ Dyspnea may be delayed and may occur mainly after exertion.

ABG = arterial blood gas; BNP = brain (B-type) natriuretic peptide; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; DVT = deep venous thrombosis; ECG = electrocardiography; MI = myocardial infarction; S3 = third heart sound; URI = upper respiratory infection; V/Q = ventilation/perfusion.

TABLE

Some Causes of Subacute* Dyspnea

Cause	Suggestive Findings	Diagnostic Approach†
Pulmonary causes		
<u>COPD</u> exacerbation	Cough, productive or nonproductive	History and physical examination
	Poor air movement Accessory muscle use or pursed lip breathing	Sometimes chest radiograph and arterial blood gas measurement
<u>Pneumonia</u>	Fever, productive cough, dyspnea, sometimes pleuritic chest pain	Chest radiograph Sometimes blood and sputum cultures
	Focal lung findings, including crackles, decreased breath sounds, and egophony	Complete blood count with differential
Cardiac causes		
<u>Angina</u> or <u>CAD</u>	Substernal chest pressure with or without radiation to the arm or jaw, often provoked by physical exertion, particularly in	ECG Cardiac biomarkers (eg, troponin) Cardiac stress testing Cardiac catheterization

Cause	Suggestive Findings	Diagnostic Approach†
	patients with risk factors for CAD	
<u>Heart failure</u>	Crackles, S3 gallop, and signs of central or peripheral volume overload (eg, elevated neck veins, peripheral edema) Orthopnea or paroxysmal nocturnal dyspnea	Chest radiograph BNP measurement Echocardiography
<u>Pericardial effusion</u> or <u>tamponade</u>	Muffled heart sounds or enlarged cardiac silhouette in patients with risk factors for pericardial effusion (eg, cancer, pericarditis, SLE) Possibly pulsus paradoxus Orthopnea	Echocardiography
Other causes		
<u>Metabolic acidosis</u>	Tachypnea	Basic metabolic panel ABG measurement
<u>Hyperthyroidism</u>	Tachycardia, warm skin, lid lag and exophthalmos (thyroid eye disease), tremor	TSH and free T4 measurement
* Subacute dyspnea occurs within hours or days. † Most patients should have pulse oximetry and, unless symptoms are clearly a mild exacerbation of a known chronic disease, chest radiograph. ABG = arterial blood gas; BNP = brain (B-type) natriuretic peptide; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; ECG = electrocardiogram; SLE = systemic lupus erythematosus; S3 = third heart sound; T4 = thyroxine; TSH = thyroid stimulating hormone.		

TABLE

Some Causes of Chronic* Dyspnea

Cause	Suggestive Findings	Diagnostic Approach†
Pulmonary causes		
<u>Interstitial lung disease</u>	Progressive dyspnea and cough Occupational and environmental exposure (occupational lung disease such as silicosis, hypersensitivity pneumonitis) may be present Smoking history, family history, medication history, or radiation history may be present Fine crackles, frequently accompanied by dry cough Clubbing	High-resolution chest CT Pulmonary function testing
<u>Obstructive lung disease</u>	Extensive smoking history, barrel chest, and poor air entry and exit or wheezing (COPD) Wheeze, cough, chest tightness, episodic symptoms, characteristic triggers (asthma)	Chest radiograph Pulmonary function testing (at initial evaluation)
<u>Pleural effusion</u>	Pleuritic chest pain, lung field that is dull to percussion and has diminished breath sounds Sometimes history of cancer, heart failure (due to increased pulmonary capillary pressure), rheumatoid arthritis,	Chest radiograph Often chest CT and thoracentesis

Cause	Suggestive Findings	Diagnostic Approach†
	systemic lupus erythematosus, or acute pneumonia	
Neuromuscular disease	Progressive dyspnea in patients with a known neuromuscular impairment (eg amyotrophic lateral sclerosis or muscular dystrophies)	Pulmonary function testing EMG or nerve conduction testing Sometimes blood tests to determine the cause
Cardiac causes		
Heart failure	Crackles, S3 gallop, and signs of central or peripheral volume overload (eg, elevated neck veins, peripheral edema) Orthopnea or paroxysmal nocturnal dyspnea	Chest radiograph Echocardiography
Pulmonary hypertension	Loud S2, parasternal heave, elevated jugular venous pressure, murmur of tricuspid regurgitation	Echocardiography Right heart catheterization
Stable angina or CAD	Substernal chest pressure with or without radiation to the arm or jaw, often provoked by physical exertion, particularly in patients with risk factors for CAD	ECG Cardiac stress testing Sometimes cardiac catheterization
Other causes		
Anemia	Normal lung examination and pulse oximetry measurement Sometimes systolic heart murmur due to increased flow	Complete blood count

Cause	Suggestive Findings	Diagnostic Approach†
Physical deconditioning	Dyspnea only on exertion in patients with sedentary lifestyle	History and physical examination
* Chronic dyspnea that has lasted for weeks to years. In many of these disorders, dyspnea is initially present on exertion before progressing to dyspnea at rest. † Most patients should have pulse oximetry and, unless symptoms are clearly a mild exacerbation of a known chronic disease, chest radiograph. CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; CT = computed tomography; ECG = electrocardiogram; EMG = electromyography; S2 = second heart sound; S3 = third heart sound.		

Etiology reference

1. [Pratter MR, Curley FJ, Dubois J, Irwin RS](#): Cause and evaluation of chronic dyspnea in a pulmonary disease clinic. *Arch Intern Med* 149 (10): 2277–2282, 1989.

Pathophysiology of Dyspnea

Although dyspnea is a relatively common problem, the pathophysiology of the uncomfortable sensation of breathing is poorly understood. Unlike those for other types of noxious stimuli, there are no specialized dyspnea receptors (although MRI studies have identified a few specific areas in the midbrain that may mediate perception of dyspnea).

The experience of dyspnea likely results from a complex interaction between chemoreceptor stimulation, mechanical abnormalities in breathing, abnormalities in cardiac function or venous return, and the perception of those abnormalities by the central nervous system. Some authors have described the imbalance between neurologic stimulation and mechanical changes in the lungs and chest wall as neuromechanical uncoupling ([1](#), [2](#)).

Pathophysiology references

1. [Mendonca CT, Schaeffer MR, Riley P, Jensen D](#): Physiological mechanisms of dyspnea during exercise with external thoracic restriction: role of increased neural respiratory drive. *J Appl Physiol* (1985) 116(5):570-581, 2014. doi:10.1152/japplphysiol.00950.2013
2. [Schwartzstein RM, Campbell ML](#). Dyspnea and Mechanical Ventilation: The Emperor Has No Clothes. *Am J Respir Crit Care Med* 205(8):864-865, 2022. doi:10.1164/rccm.202201-0078ED

Evaluation of Dyspnea

History

History of present illness should cover the temporal onset (eg, abrupt, insidious) and duration.

Severity can be determined by assessing the activity level required to cause dyspnea (eg, dyspnea at rest is typically more severe than dyspnea only when climbing stairs). Healthcare professionals should note how much dyspnea has changed from the patient's usual state. Precipitating or exacerbating factors (eg, allergen exposure, cold, exertion, supine position) and relieving factors should be identified.

Review of systems should seek symptoms of possible causes, including chest pain or pressure ([pulmonary embolism](#), myocardial ischemia, [pneumonia](#)); dependent edema, orthopnea, and paroxysmal nocturnal dyspnea ([heart failure](#)); fever, chills, cough, and sputum production (pneumonia); black, tarry stools or heavy menses (occult bleeding, possibly causing [anemia](#)); and weight loss or night sweats (cancer or chronic lung infection).

Past medical history should cover disorders known to cause dyspnea, including [asthma](#), [COPD](#), and heart disease, as well as risk factors for the different etiologies:

- Smoking history: For cancer, COPD, some [interstitial lung diseases](#), and heart disease
- Family history, hypertension, and high cholesterol levels: For [coronary artery disease](#)
- Recent immobilization or surgery, recent long-distance travel, cancer or risk factors for or signs of occult cancer, prior or family history of clotting, pregnancy, oral contraceptive use, calf pain, leg swelling, and known deep venous thrombosis: For [pulmonary embolism](#)

Environmental and occupational exposures (eg, [gases](#), [smoke](#), [asbestos](#)) should be investigated.

Physical examination

Vital signs are reviewed for fever, tachycardia, hypoxia, and tachypnea.

Examination focuses on the cardiovascular and pulmonary systems.

A full lung examination should be performed, particularly including adequacy of air entry and exit, symmetry of breath sounds, and presence of crackles, rhonchi, stridor, and wheezing. Signs of consolidation (eg, egophony, dullness to percussion) should be sought. The cervical, supraclavicular, and inguinal areas should be inspected and palpated for lymphadenopathy.

Neck veins should be inspected for distention, and the legs and presacral area should be palpated for pitting edema (both suggesting [heart failure](#)).

Heart sounds should be auscultated with notation of any extra heart sounds, muffled heart sounds, or murmurs. Testing for [pulsus paradoxus](#) (a > 12-mm Hg drop of systolic blood pressure during inspiration) can be done by inflating a blood pressure cuff to 20 mm Hg above the systolic pressure and then slowly deflating until the first Korotkoff sound is heard only during expiration. As the cuff is further deflated, the point at which the first Korotkoff sound is audible during both inspiration and expiration is

recorded. If the difference between the first and second measurement is > 12 mm Hg, then pulsus paradoxus is present.

Conjunctiva should be examined for pallor.

Red flags

The following findings are of particular concern:

- Dyspnea at rest during examination
- Decreased level of consciousness or agitation or confusion
- Accessory muscle use and poor air excursion
- Chest pain
- Crackles
- Weight loss
- Night sweats
- Palpitations

Interpretation of findings

The history and physical examination often suggest a cause and guide further testing (see tables [Some Causes of Acute Dyspnea](#), [Some Causes of Subacute Dyspnea](#), and [Some Causes of Chronic Dyspnea](#)) (1). Several findings are of note:

- [Wheezing](#) suggests [asthma](#) or [COPD](#), although it can occur with other disease processes such as [heart failure](#), [paradoxical vocal fold motion](#), [tracheal stenosis](#), tracheomalacia, [relapsing polychondritis](#), and [bronchiolitis](#).
- [Stridor](#) suggests extrathoracic airway obstruction (eg, foreign body, [epiglottitis](#), [vocal fold dysfunction](#)).
- Crackles suggest left [heart failure](#), [interstitial lung disease](#), or, if accompanied by signs of consolidation, [pneumonia](#).

However, the symptoms and signs of life-threatening conditions such as myocardial ischemia and pulmonary embolism can be nonspecific. Furthermore, the severity of symptoms is not always proportional to the severity of the cause (eg, pulmonary embolism in a fit, healthy person may cause only mild dyspnea). Thus, a high degree of suspicion for these common conditions is prudent. It is often appropriate to exclude these conditions before attributing dyspnea to a less serious etiology.

Consideration of pre-test probability and sometimes clinical prediction rules can help estimate the likelihood of [pulmonary embolism](#). Note that normal oxygen saturation does not exclude pulmonary embolism.

[Hyperventilation syndrome](#) is a diagnosis of exclusion. Because hypoxia may also cause tachypnea and agitation, it should not be assumed that every rapidly breathing, anxious young person merely has hyperventilation syndrome.

Testing

Pulse oximetry should be performed in all patients, and chest radiography should be done as well unless symptoms are clearly caused by a mild or moderate exacerbation of a known condition. For example, patients with asthma or heart failure do not require a radiograph for each flare-up, unless clinical findings suggest another cause or an unusually severe attack.

Most adults should have an ECG to detect myocardial ischemia (and serum cardiac biomarker testing if suspicion is high) unless myocardial ischemia can be excluded clinically.

In patients with severe or deteriorating respiratory status, arterial blood gases (ABGs) should be measured to more precisely quantify PaO₂ and PaCO₂, diagnose [acid-base disorders](#) stimulating hyperventilation, and calculate the alveolar-arterial oxygen gradient. Also, obtaining a basic metabolic panel to measure the bicarbonate level can be useful to assess the chronicity of a [respiratory acidosis](#).

Patients who have no clear diagnosis after chest radiograph and ECG and are at moderate or high risk of having pulmonary embolism (based on a clinical prediction rule) should undergo CT angiography or ventilation/perfusion scanning. Patients who are at low risk may have D-dimer testing (a normal D-dimer level effectively excludes pulmonary embolism in a low-risk patient). See diagnostic criteria to determine the likelihood of a patient having a [pulmonary embolism](#).

Chronic dyspnea may warrant additional tests, such as CT, pulmonary function tests, echocardiography, and bronchoscopy. If initial workup is inconclusive, [cardiopulmonary exercise testing](#) can be considered.

CLINICAL CALCULATORS

[A-a Gradient](#)



Evaluation reference

1. [Parshall MB, Schwartzstein RM, Adams L, et al](#): An Official American Thoracic Society Statement: Update on the mechanisms, assessment, and management of dyspnea. *Am J Respir Crit Care Med* 185:435–452, 2012.

Treatment of Dyspnea

Treatment is correction of the underlying disorder.

Hypoxemia (in the absence of compensatory hyperventilation) is treated with supplemental oxygen as needed to maintain oxygen saturation > 88% or PaO₂ > 55 mm Hg (> 7.3 kPa) because levels above these thresholds provide adequate oxygen delivery to tissues. Levels below these thresholds are on the steep portion of the oxygen–hemoglobin dissociation curve, where even a small decline in arterial oxygen tension can result in a large decline in hemoglobin saturation. Oxygen saturation should be maintained at > 93% if myocardial or cerebral ischemia is a concern, although data suggest that

supplemental oxygen is not beneficial in the treatment of acute myocardial infarction unless the patient has hypoxia (1).

Morphine (eg, 0.5 to 5 mg IV) helps reduce anxiety and the discomfort of dyspnea in various conditions, including myocardial infarction, pulmonary embolism, and the dyspnea that commonly accompanies terminal illness. However, opioids should be used with caution in patients with acute airflow limitation (eg, asthma, COPD) because they suppress the ventilatory drive and can worsen respiratory acidemia.

Treatment reference

1. [Cabello JB, Burls A, Emparanza JJ, Bayliss SE, Quinn T](#): Oxygen therapy for acute myocardial infarction. *Cochrane Database Syst Rev* 12(12):CD007160, 2016.
doi:10.1002/14651858.CD007160.pub4

Key Points

- Pulse oximetry is a key component of the examination.
- Low oxygen saturation (< 90%) indicates a serious condition, but normal saturation does not exclude one.
- Accessory muscle use, a sudden decrease in oxygen saturation, or a decreased level of consciousness requires emergent evaluation and hospitalization.
- Myocardial ischemia and pulmonary embolism are relatively common, but their symptoms and signs can be nonspecific.
- Exacerbation of known conditions (eg, asthma, chronic obstructive pulmonary disease, heart failure) is common, but patients may also develop new conditions.

Drug Information for the Topic



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